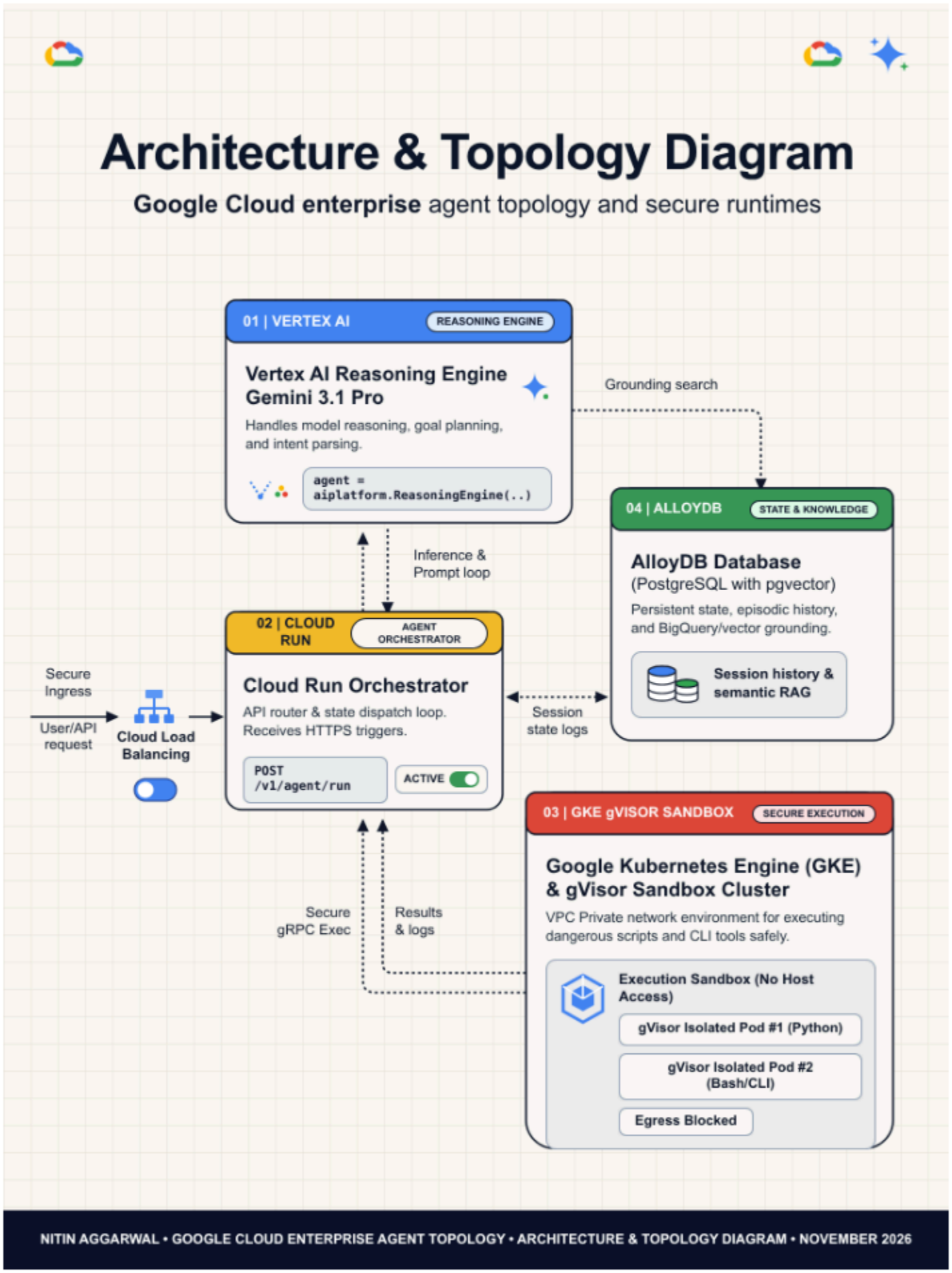
Anti-'Context Rot' L1/L2/L3 Cache & URI Pointer Harness

CPU L1/L2/RAM Virtual Memory Hierarchy vs. Stuffing Every Raw Log File into CPU Registers Pin immutable rules at token 0 for warm KV-prefix hits and replace >2KB tool outputs with typed `artifact://` URI pointers.



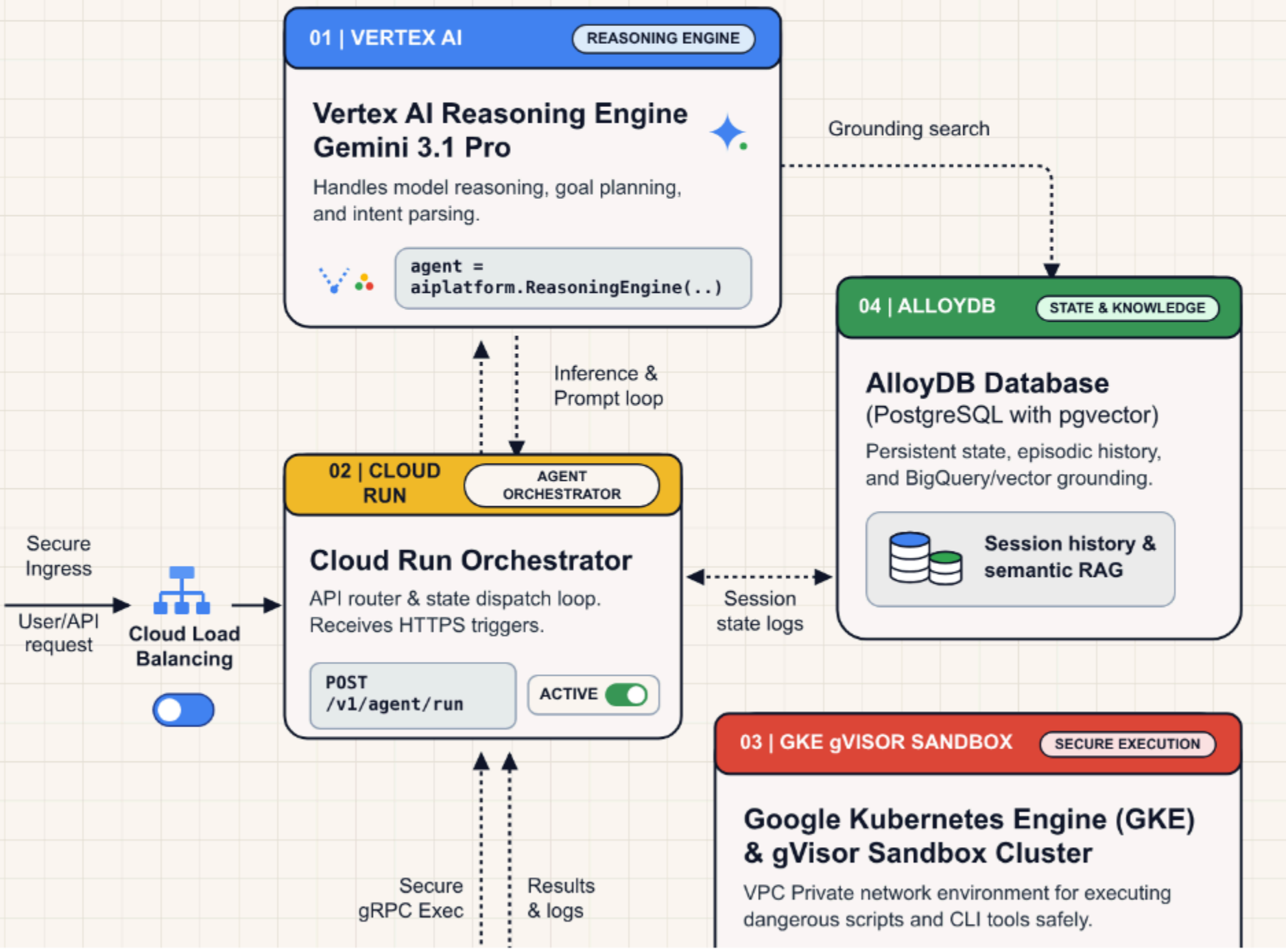
- **Engineering Pillar: MCP, Tool Routing & Zero-Trust Identity**
- **Verified Impact: 94.8% KV cache hit rate, 9.4s -> 0.9s TTFT (10.4x), 97.1% Turn-50 instruction adherence.**
- **Dual Stack: Vertex AI Gemini (locations/global) • Cloud Run ADK • Model Armor**

Why Naive Agent Loops Fail in Enterprise Production

Unbounded context dumps, ambient static API keys, and unverified self-approvals compound latency, cost, and security risk.

Architecture & Topology Diagram

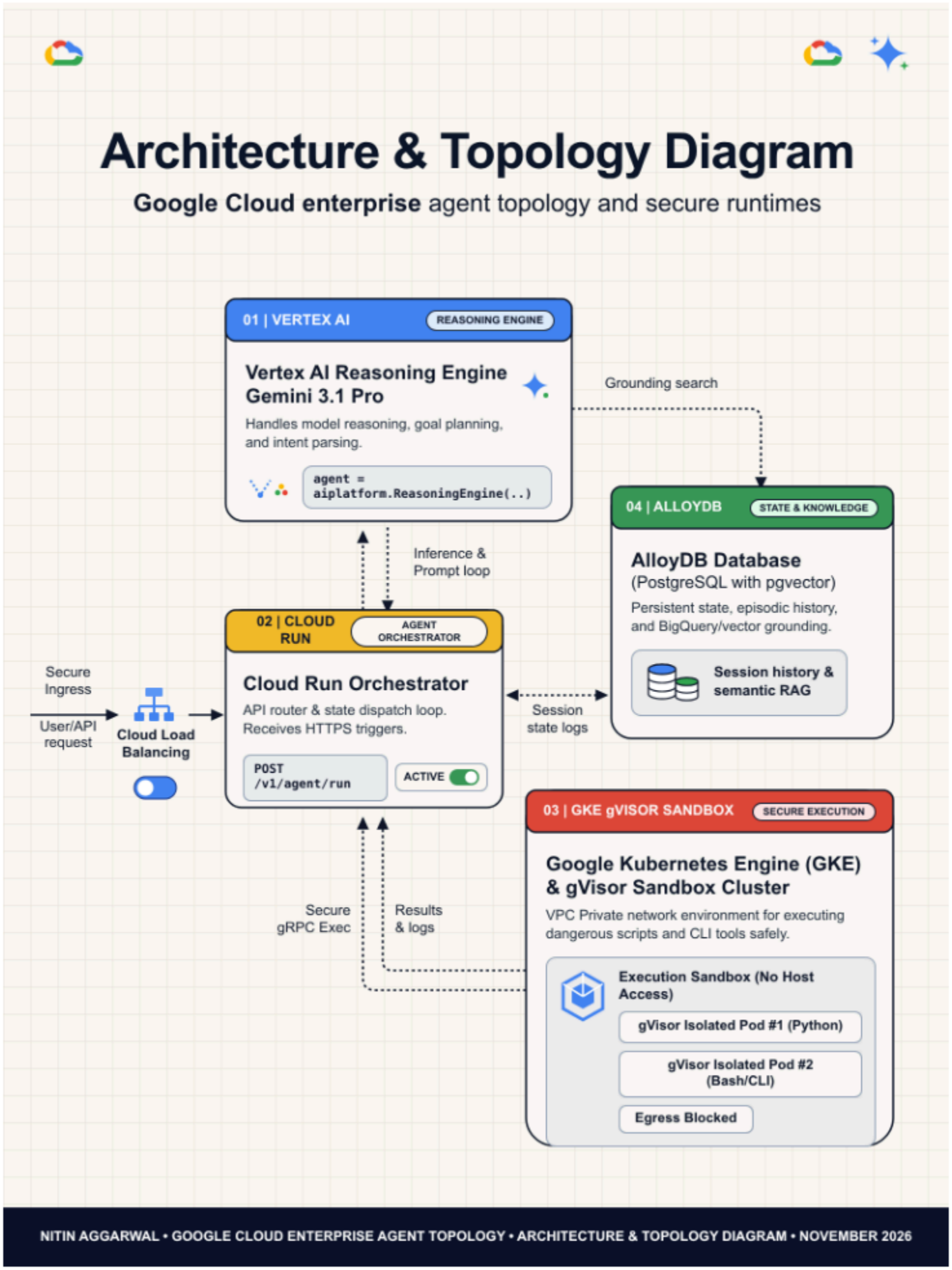
Google Cloud enterprise agent topology and secure runtimes



- ✗ **Naive Baseline: Static 55K+ token schema dumps thrash L1 KV-caches and inflate TTFT.**
- ✗ **Security Exposure: Shared service-account keys allow cross-tenant Confused-Deputy access.**
- ✓ **Engineered Fix: Enforce deterministic schema budgets, narrow-audience OBO tokens, and CI replay gates outside the LLM.**

Full-Stack Reference Architecture (T61)

Grounded on Google Cloud ADK • 2026 Career-Minting Signal — pairing managed Google Cloud SLA controls with portable open-source equivalents.



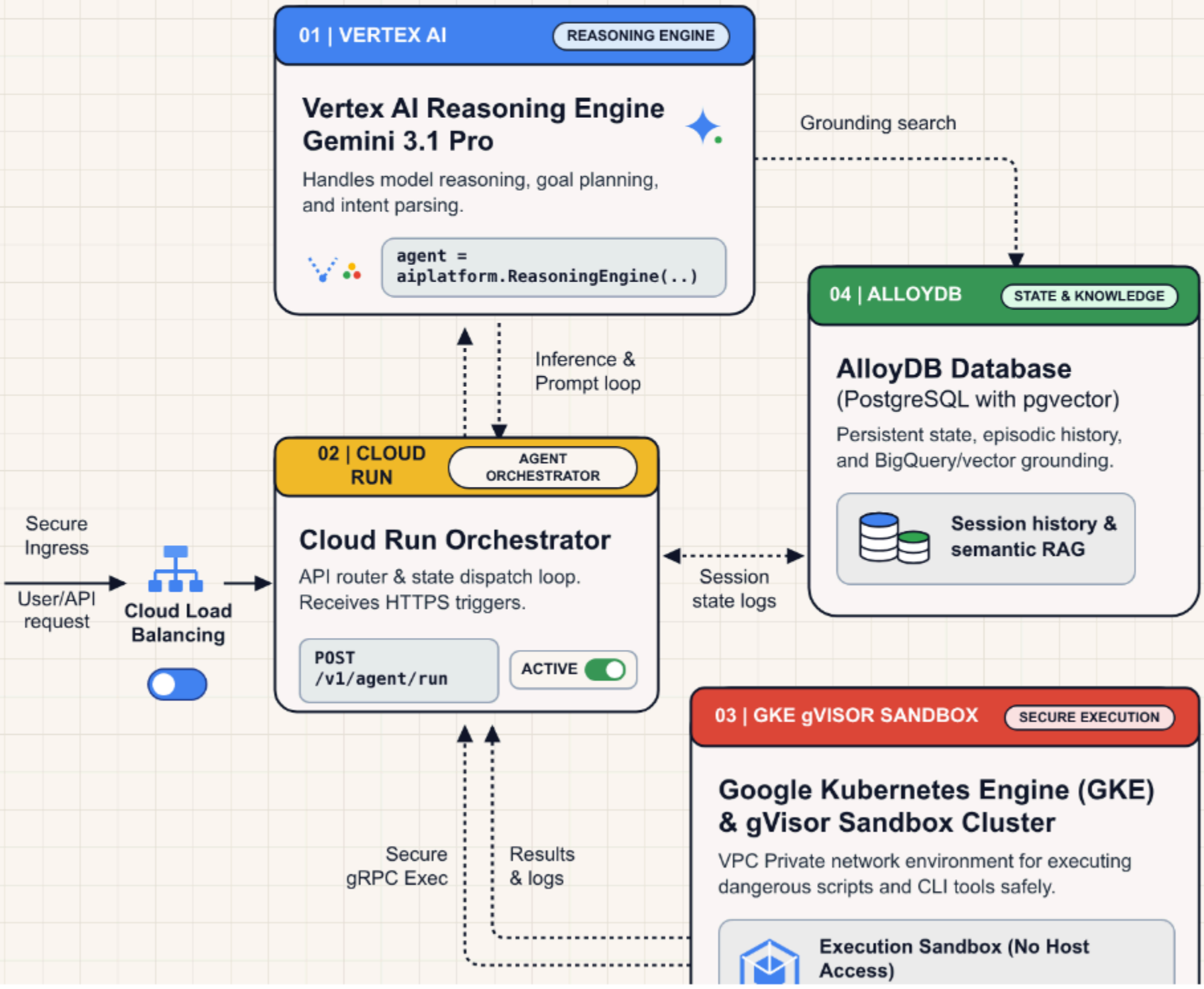
- **Google Cloud Stack: Vertex AI Gemini (locations/global) • Cloud Run ADK • Model Armor**
- **Open-Source Equivalent: OpenHands CodeAct • LangGraph / FastMCP • OpenTelemetry**

Stage 01 — Ingress, Discovery & Context Hygiene

Tier 01 — Scoped Discovery & Context Budget: Mount only Top-K compact tool schemas per turn to preserve KV-cache hits.

Architecture & Topology Diagram

Google Cloud enterprise agent topology and secure runtimes



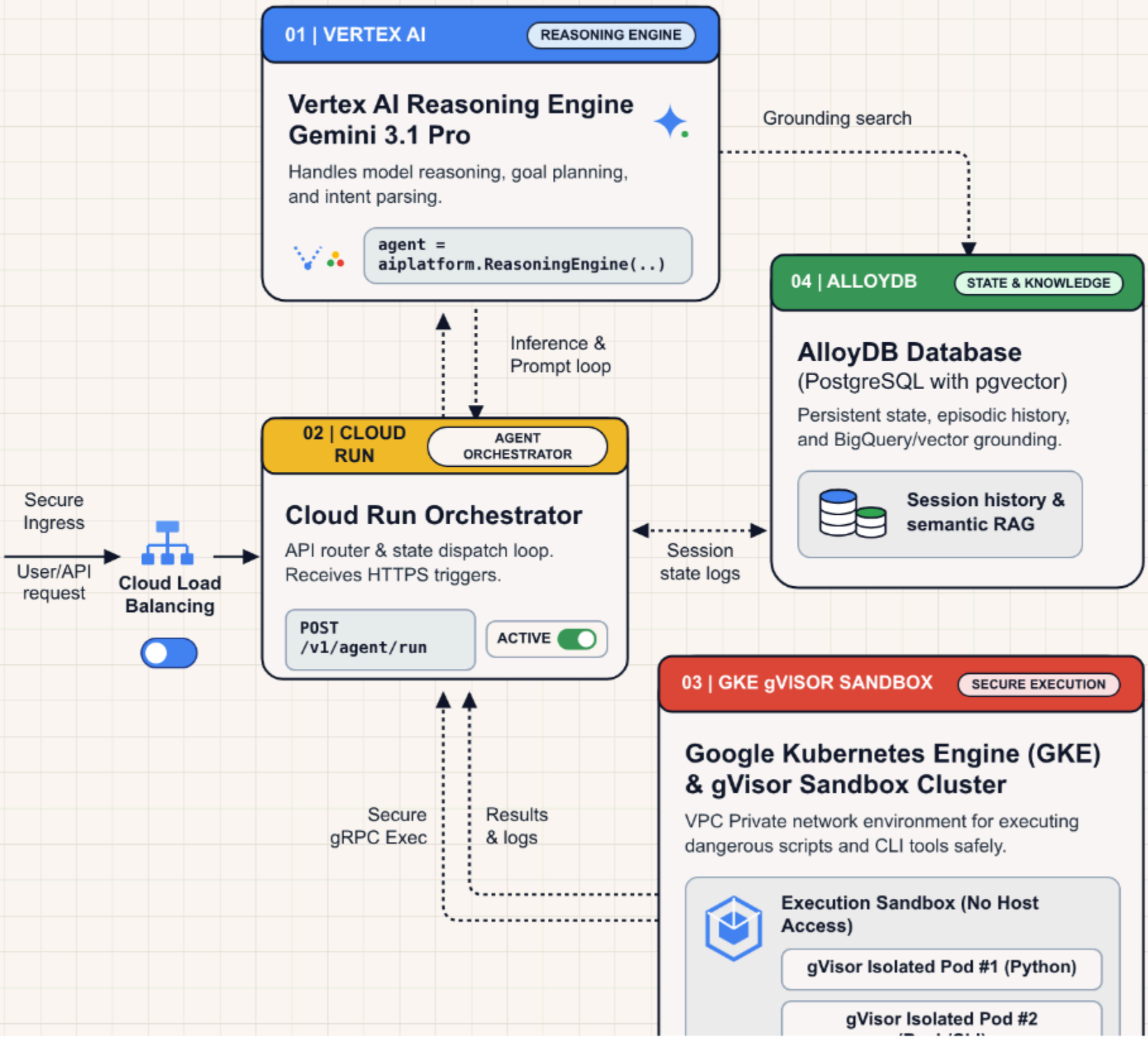
- **Pin immutable system instructions at prefix byte 0 for >90% KV-cache reuse.**
- **Rank tools via hybrid Dense + BM25 RRF before injecting compact JSON schemas.**

Stage 02 — Identity, Guardrails & Policy Enforcement

Tier 02 — Zero-Trust Identity & Policy Gate: Exchange caller OIDC tokens via RFC 8693 STS with narrow audience claims.

Architecture & Topology Diagram

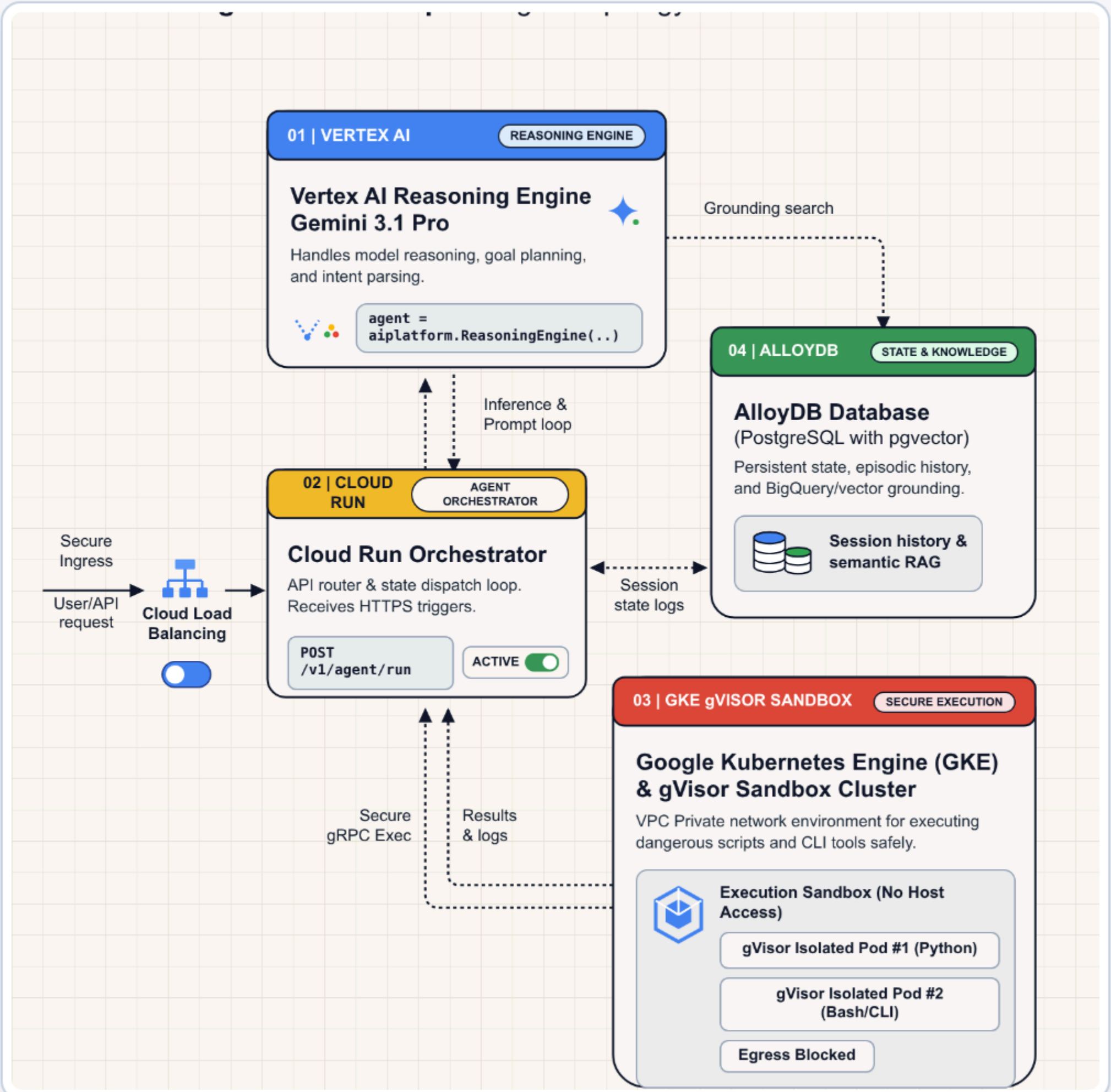
Google Cloud enterprise agent topology and secure runtimes



- **Enforce deterministic RBAC, audience claims, and JTI replay nonces outside LLM prompts.**
- **Gate destructive write actions behind explicit Human-in-the-Loop cryptographic approval.**

Stage 03 — Sandboxed Execution & Adversarial Verification

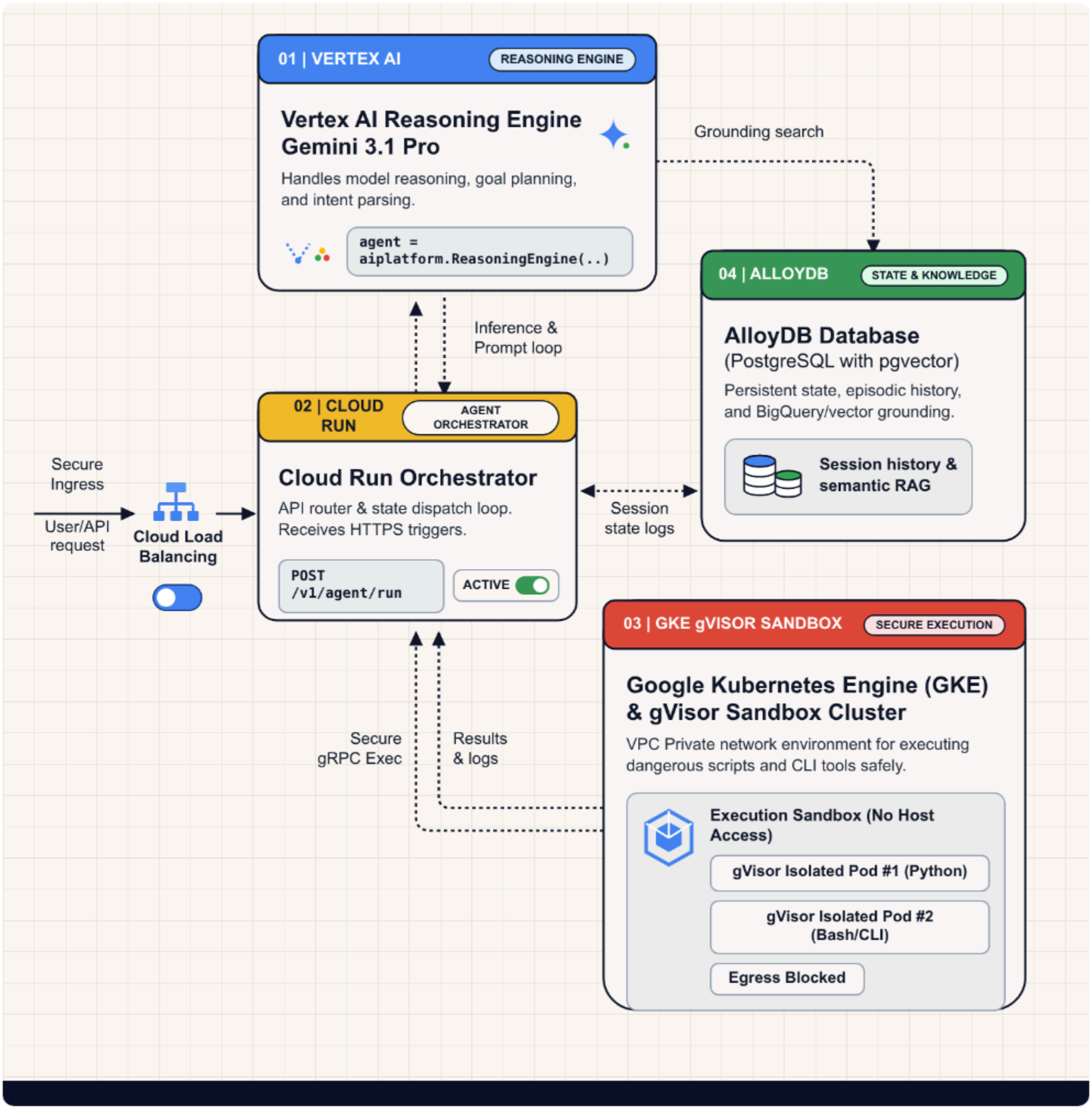
Tier 03 — Isolated Sandbox & Deterministic Verification: Validate outputs inside gVisor/OpenHands with unit and DOM gates.



- **Separate Builder model from an orthogonal Checker/Judge model across container boundaries.**
- **Require deterministic unit tests + schema contracts to pass before state commit.**

Stage 04 — Production Telemetry, SLOs & FinOps ROI

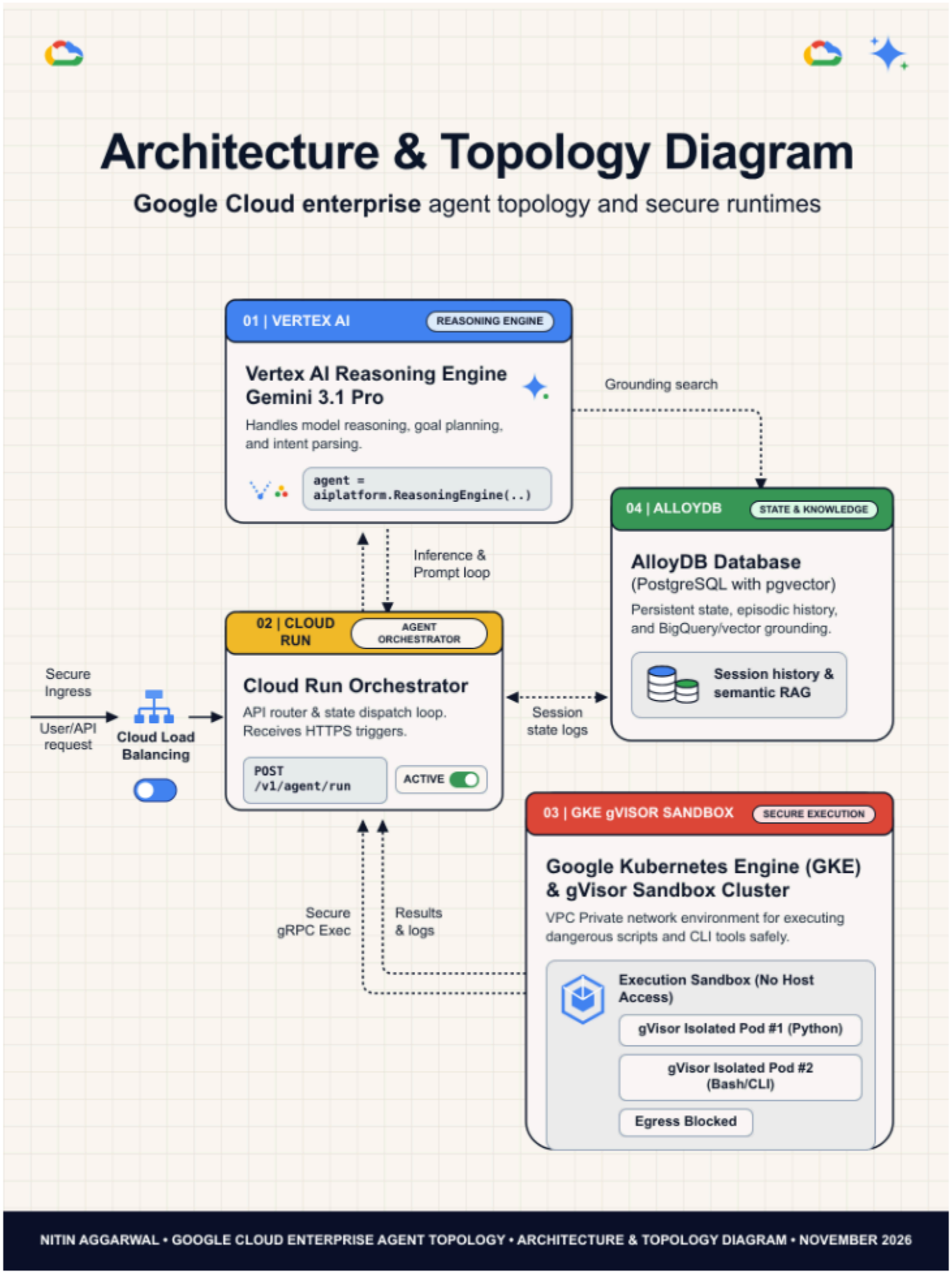
Tier 04 — Telemetry, FinOps & Replay Ledger: Persist cryptographic SHA-256 execution receipts and cost/latency budgets.



- **Measured Telemetry: 94.8% KV cache hit rate, 9.4s -> 0.9s TTFT (10.4x), 97.1% Turn-50 instruction adherence.**
- **Track p95 end-to-end latency, pass^k multi-trial stability, and cost per verified task.**

Principal Architect Golden Rule (T61)

Enforce deterministic schema budgets, narrow-audience OBO tokens, and CI replay gates outside the LLM.



- 1. Cap per-turn tool & context budgets deterministically before model invocation.
- 2. Bind narrow-scope OAuth 2.1 OBO identity + Model Armor policy gates.
- 3. Gate every deployment on a seeded multi-trial replay benchmark.